MILITARY SPECIFICATION
SWITCH, THERMOSTATIC, MARK 3 MOD 0

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1 1 Scope. This specification establishes the requirements for the procurement of the Thermostatic Switch Mark 3 Mod 0. The thermostatic switch is of one type and one class.

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

QQ-C-320	Chromium plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)

MILITARY

MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

STANDARDS

MILITARY

MS 24347	Drums, Metal, Reusable Shipping and Storage
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Test for

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

LD 496636	Container, Inner, for Thermostatic Switch Mark 3 Type
LD 496637	Container, Outer, for Thermostatic Switch Mark 3 Type

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

WR-43	Naval Weapons Requirements, Preparation of Quality Assurance Requirements

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications The following document forms a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 Screw Thread Standards for Federal Services.

(Copies are available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402)

3 REQUIREMENTS

3 1 Description The thermostatic switch, hereinafter referred to as "the switch", is a single-throw, single-pole, normally-closed, thermal-responsive switch of a fixed-temperature, snap-action type, requiring snap action upon closure only. (See 6.3 1) The contacts of the switch are enclosed within a corrosion-resistant housing that is centrally arranged and fastened to a mounting plate by means of which the switch is secured in service. The overall dimensions of the switch, the location of the mounting holes in the mounting plate, the terminal screw finish, and descriptive marking are as shown on Figure 1 of this specification.

3.2 First article sample. Unless otherwise specified in the contract or order, a first article sample of the switch shall be manufactured using the methods and procedures proposed for the production lot. (See 6.2) The sample will be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of producing items that comply with this specification. The sample size shall be as specified in the contract or purchase order

3.3 General requirements.

3 3 1 Materials Unless otherwise specified in the contract, the materials used in the manufacture of the switch shall be in accordance with Figure 1. (See 6.2)

3.3.2 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the switch shall be in accordance with Figure 1.

3 3 3 Marking Marking shall be in accordance with Figure 1.

3.3 4 Interchangeability Unless otherwise specified, component parts of assembled units shall be interchangeable and shall assemble with mating components without selection or fitting.

3.3.5 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28

3 4 Performance requirements and product characteristics The switch shall have the following performance requirements and product characteristics

3.4.1 Contact force The movable contact of the switch shall resist a 60-gram force (0.59 N) (See 6 4) applied in a direction to separate the contacts, without increasing the resistance between the contacts to a value exceeding the contact resistance requirements

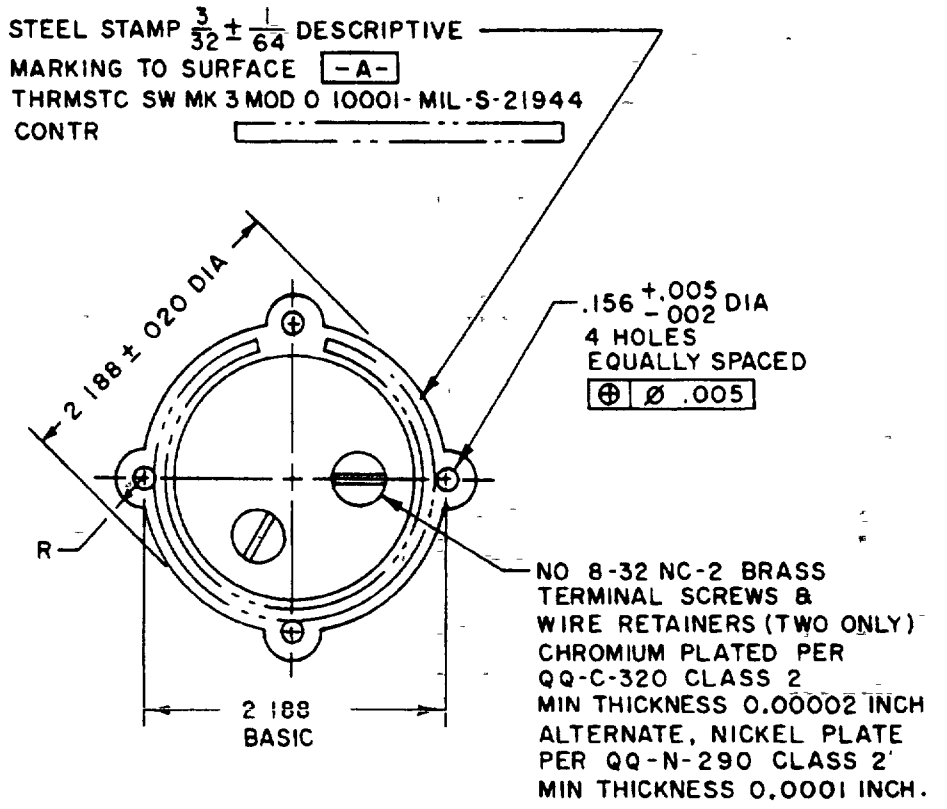
3.4.2 Contact resistance At any ambient temperature between 65° and 95°F (18° and 35°C), the resistance between the closed contacts of the switch, when measured at the switch terminals, shall not exceed 0.2 ohm.

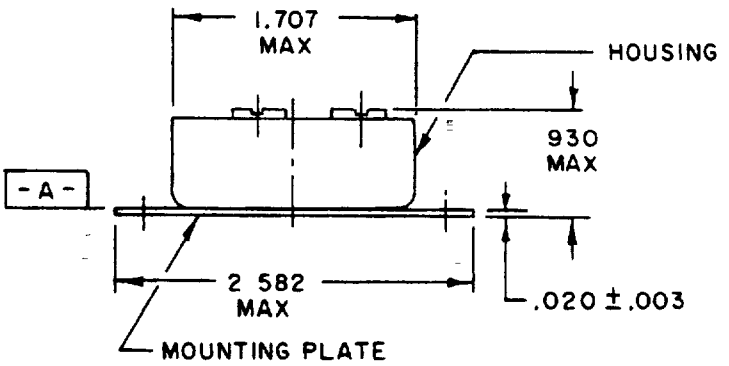
3.4.3 Insulation resistance The insulation resistance between either terminal of the switch and the housing shall be 50 megohms or greater.

3 4 4 Insulation resistance (low temperature) With the temperature of the switch reduced sufficiently to open the contacts, the insulation resistance between the contacts and between each of the contacts and the switch housing shall be 50 megohms or greater

3 4 5 Temperature response The contacts of the switch shall be open at 2°F (-17°C) and have the required insulation resistance The contacts of the switch shall close during an increase in temperature from 2° to 28°F (-17° to -2°C) and have the required contact resistance after closure

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.





INCHES	MM
0 00002	0 0005
0 0001	0 0025
0 002	0 050
0 003	0 075
0 005	0 125
1/64	0.40
0 020	0 510
3/32	2.40
0 156	3 960
0 930	23 60
1 707	43 35
2 188	55.60
2 582	65.60

Notes 1. Dimensions are given in inches.
2 Inches to mm table for information only, not exact equivalents

Figure 1 Thermostatic Switch-Configuration and Dimensions

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

3 5 Environmental requirements

3.5 1 Vibration. The switch shall meet all performance requirements and product characteristics of this specification after being vibrated in accordance with 4.6.1.

3.5.2 Shock. The switch shall meet all performance requirements and product characteristics of this specification after being shocked in accordance with 4.6.2.

3.5.3 Storage. The switch shall meet all performance requirements and product characteristics of this specification after being subjected to one 14-day JAN TEMPERATURE AND HUMIDITY CYCLE defined in MIL-STD-331, Test Method 105, except that the relative humidity in the chamber need not exceed room ambient or ambient as controlled

3.6 Clean-up. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of metal chips and other foreign matter. Burrs and sharp edges shall be removed

3 7 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6 2)

4.2 Classification of inspections. Inspection of the switches shall be classified as follows

a First article inspection (4.3.1)
b. Quality conformance inspection (4.3.2)
c Periodic production inspection (4.3 3)

4.3 Inspections

4.3.1 First article inspection. Each first article sample shall be subjected to first article inspection in accordance with Table I. The failure of a sample to conform to any requirement of this specification shall be cause for denial of production approval.

4 3.2 Quality conformance inspection Each switch shall be subjected to quality conformance inspection in accordance with Table I. (100% Inspection) The failure of a switch to conform to any requirement of this specification shall be cause for rejection of that switch

4.3.3 Periodic production inspection A sample shall be drawn from each production lot of switches for periodic production inspection. (See 6.2, 6.3.2) The sample size and method of selecting the sample shall be as specified in the contract or purchase order. (See 6.2) Each switch in the sample shall be subjected to periodic production inspection in accordance with Table I The failure of a sample switch to conform to any requirement of this specification shall be cause for rejection of the sample and the lot it represents.

4.3.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Major or Minor in accordance with WR-43. Tests and examinations that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination Tests and examinations which are not annotated with a classification code are classified Minor, AQL 2.5 percent

4 4 Test equipment The following items of test equipment are required to perform the acceptance tests set forth in this specification

a. A temperature conditioning chamber capable of cooling to 0°F (-18°C)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

b A megohmmeter having a 500 ± 50 volts DC test potential and capable of measuring at least 50 megohms with an accuracy within 5 percent.

c. A resistance measuring device or meter capable of measuring 0.1 to 1 0 ohm with an accuracy of one percent.

d A metal Drum MS 24347-26 with Locking Ring MS 24347-63 and Gasket MS 24347-83.

e A Metal Drum Lid MS 24347-43 modified with tip jacks installed so that thermostatic switch operation can be monitored while maintaining airtight integrity of the drum

f A means of monitoring the temperature inside the drum.

g. Eight-unit desiccant bags in accordance with MIL-D-3464 as required.

TABLE I - INSPECTION PROCEDURES

Title of Examination or Test	Requirement Paragraph	Method Paragraph	INSPECTION CLASSIFICATIONS		
			First Article (1)	Quality Conformance	Periodic Production (1)
Preparation for Delivery	5	4.5.1	1	(2)	1
Materials, construction, dimensions, marking, threads, clean-up, and workmanship	3.3.1, 3.3.2, 3.3.3, 3.3.5, 3.6, and 3.7	4.5.2 and 4.5.3	2	(2)	2
Contact resistance	3.4.2	4.5.5	3	(2)	3
Insulation resistance	3.4.3	4.5.6	4	(2)	4
Temperature response	3 4.5	4.5.8.1 and 4.5 8 2	5	(2)	5
Insulation resistance (low temperature)	3.4.4	4.5.7	6 (3)	(2)	6 (3)
Vibration	3 5 1	4.6.1	7 (4)		7 (4)
Shock	3 5 2	4 6.2	8 (4)		8 (4)
Storage	3.5.3	4.6.3	9 (4)		9 (4)
Contact force	3 4 1	4 5.4	10 (5)	(2)	10 (5)

(1) These inspections shall be performed in successive numerical order, unless otherwise specified.
(2) Quality conformance inspection shall be an in-process function performed during and after assembly of the switch prior to delivery by the manufacturer. The sequence shall be at the manufacturer's discretion.
(3) This test may be performed during the tests of 4.5.8.
(4) Repeat steps 3, 4, 5, and 6.
(5) The quantities of switches to be subjected to this test shall be as determined by the testing activity

4.5 General examination and test methods (See Table I.)

4 5.1 Preparation for delivery. The packaging shall be examined to determine conformance to 5 herein

4 5 2 General examination. Each switch shall be visually and dimensionally examined to determine conformance to Figure 1 and 3.3 herein

4.5.3 Cleanliness examination The switch and its components shall be examined to determine the presence of oil, grease, dirt, dust, moisture, corrosion, metal chips, metal fillings and the like, and to determine conformance to 3.6 herein

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

4.5.4 Contact force test (M101). Support the contact assembly, terminal surface up, and connect across the terminals the Ohmmeter, Item 4.4c. Apply not less than a 60-gram force (0.59 N) (See 6.4) against the arm of the movable contact as close as possible to the contact and in a direction to open the contacts. With the force applied against the movable contact, verify that the contact resistance does not exceed 0.2 ohm

4.5.5 Contact resistance test (M102) At an ambient temperature between 65° and 95°F (18° and 35°C), connect the ohmmeter, Item 4.4c., across the terminals of the switch. Verify that the contact resistance does not exceed 0.2 ohm

4.5.6 Insulation resistance test, contacts closed (M103) Place the switch in a suitable conditioning chamber for not less than 24 hours while maintaining the chamber at a temperature of 77° ± 2°F (25° ± 1°C) and a relative humidity of 75 ± 5 percent Immediately after removing the switch from the conditioning chamber, use the Megohmmeter, Item 4.4b., to measure the insulation resistance between either terminal and the housing at any temperature between 25°F and 95°F (-4° and 35°C) and a relative humidity of 60 percent or greater. Verify that the insulation resistance is not less than 50 megohms.

4.5.7 Insulation resistance test, contacts open (M104). Reduce the temperature of the switch sufficiently to open the contacts. At this temperature, use the Megohmmeter, Item 4.4b, to measure the insulation resistance between the two switch terminals and between each switch terminal and the housing. Verify that the insulation resistance is not less than 50 megohms in all measurements

4.5.8 Tests for operating temperature (M105) Perform the following test on the completed located in the Metal Drum, Item 4.4d., containing one desiccant bag. Monitor the switch temperature and operation at the electrical tip jacks installed in the drum lid

4.5.8 1 Switch-opening temperature test. Condition the switch in the drum containing a fresh air dryer for not less than 0.5 hour at ambient temperature. Lower the temperature in the drum. When the temperature inside the drum reaches 2°F (-17°C), subject the switch to the insulation resistance test of 4.5.7. An insulation resistance in excess of 50 megohms indicates that the switch is open.

4.5.8.2 Switch-closing temperature test With the switch contacts open, increase the temperature in the drum When the temperature inside the drum reaches 28°F(-2°C), subject the switch to the contact resistance test of 4 5.5. A contact resistance which does not exceed 0 2 ohm indicates that the switch contacts are closed.

4.6 Environmental tests

4.6.1 Vibration The switch shall be rigidly fastened directly to the vibration table by either simulating the actual method of mounting used in service or by its external case, whichever, is appropriate. Simple harmonic excitation shall be applied parallel to each of the three principal axes of the switch for equal periods of time. The frequency range from 10 to 60 Hz may be covered either by slowly cycling at a logarithmic rate between the limits, or by at least 24 discrete frequency steps which have a logarithmic distribution. The vibration amplitude shall be maintained at 2 ± 2 gravity units (g) or 0.10 ± .01 inch (2.54 ± 0.25mm) (peak-to-peak), whichever is the lesser The duration of the test in each axis shall be not less than four hours. The total test duration shall be not less than twelve hours.

4.6.2 Shock. The switch shall be subjected to a continuous two-phase shock applied in any direction with respect to the switch

a The first (impact) phase shall consist of a change of velocity of 5.5 to 7 0 feet per second (1.68 to 2.13m/s) (See 6.4) taking place in from 0.2 to 0.4 millisecond.

b. During the second (drag) phase, the switch shall be accelerated for a minimum time of 15 milliseconds in such a manner that during this time the minimum average acceleration is 90g The peak acceleration shall be between 125 and 150 g.

4 6 3 Storage temperature. Subject the switch, packaged and packed in accordance with 5, to one 14-day JAN TEMPERATURE AND HUMIDITY CYCLE defined in MIL-STD-331, Test Method 105, except that the relative humidity in the chamber need not be raised above room ambient as controlled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

5 PREPARATION FOR DELIVERY (see 6.2)

5.1 Preservation and packaging. Preservation and packaging shall be Level A. The switches shall be packaged in inner containers in accordance with LD 496636.

5.2 Packing Packing shall be Level A Inner containers shall be packed in accordance with LD 496637

5.3 Marking In addition to any special marking required by the contract or order, all marking shall be in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The thermostatic switch is intended for use in underwater mines

6.2 Ordering data. Procurement documents should specify the following

a Title, number, and date of this specification.
b Number of switches required for the first article sample. (3.2)
c That the standards of workmanship exhibited by previously approved production sample shall be used to determine the minimum requirements of the current contract relative to workmanship, when first article sample requirements are waived because of having been submitted on a recent contract. (3.7)
d. Exceptions and additions to the requirements of Figure 1 (3.3.1)
e Activity to perform first article sample inspection, if other than contractor. (4.1)
f. Activity to perform periodic production sample inspection, if other than contractor. (4.1)
g. Lot size, if one month's scheduled production is not acceptable (4.3.3)
h The sample size and selection procedure to be used for periodic production inspection. (4.3.3)
i Test equipment to be loaned by the Government, and the terms under which it is loaned. (4.4)
j. That preparation for delivery shall be adaquate to protect the switch prior to its installation, when switches are purchased for a prime contractor for installation into Assembly Condition "D" mines (5)

6.3 Definitions The following terms are used throughout this specification and are here defined for clarity

6.3.1 Thermostatic switch Reference herein to the switch shall be understood to mean the Thermostatic Switch Mark 3 Mod 0.

6.3 2 Lot Unless otherwise specified in the contract or purchase order, lot size shall mean one month's scheduled production. See Section 5 of MIL-STD-105 for lot composition.

6.4 Metrication. This document has been annotated parenthetically with metric "equivalents" of quantities not expressed in the International System of Units (SI). Metric units are in accordance with ASTM E380-72 as adopted by the DoD on 29 November 1973, and are included because the DoD encourages conversion to metric dimensioning. Parenthetical quantities are intended as information only, as they are usually not exact equivalents, but rounded off within specified tolerances.

6.5 The design and technical cognizance for the switch specified herein are located at the Naval Mine Engineering Facility, Yorktown, Virginia 23691.

Custodian
NAVY - OS

Review Activities
NAVY - OS

Preparing Activity
NAVY - OS

Project Number
5930-N456

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ORDNANCE STATION
INDIAN HEAD, MD 20640

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DoD 316

Commanding Officer
Naval Ordnance Station
Standardization Division (502)
Indian Head, MD 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE
MIL-S-21944B(OS) "Switch, Thermostatic, Mark 3 Mod 0"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:48Z
Check the source to verify that this is the current version before use.